

		$a_c = 0.000 \text{ m s}^{-2}$	1
3	(a)	Gravitational potential at a point is the work done <u>per unit mass</u> in bringing a <u>small test mass</u> from <u>infinity to that point</u> .	1
	(b)	(i) 1. The gravitational field is non uniform. Accept: Gravitational potential varies inversely proportional to the distance from the centre of Earth to a point.	1
	(b)	(i) 2. Gravitational potential is taken to be zero at infinity. The gravitational forces are attractive and thus negative work has to be done by an external force to bring the mass from infinity to that point at constant speed.	1 1
	(b)	(ii) By Principle of Conservation of Mechanical Energy, $\text{GPE}_i + \text{KE}_i = \text{GPE}_f + \text{KE}_f$ $m\Phi_m + 0 = m\Phi_E + \frac{1}{2}mv^2 \quad [1]$ $v^2 = 2(\Phi_m - \Phi_E)$ $v = \{2[-1.0 - (-62.0)]\}^{0.5} \quad [1]$ $= 1.1 \times 10^4 \text{ m s}^{-1} \quad [1]$	
	(b)	(iii) Since gravitational potential, $\phi = -GM/r$, $-61.6 \times 10^6 = -(6.67 \times 10^{-11})(5.97 \times 10^{24}) / (6371 \times 10^3 + h)$ where h is the height of the upper edge of the atmosphere correct substitution of ϕ, G, M [1] $r = r_E + h$ [1] Thus, $h = 9.3 \times 10^4 \text{ m}$ (93.3 km) [1]	

4	(a)	The INCREASE in the internal energy of a system is equal to the sum of heat	1
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